FLAb: Benchmarking deep learning methods for antibody fitness prediction

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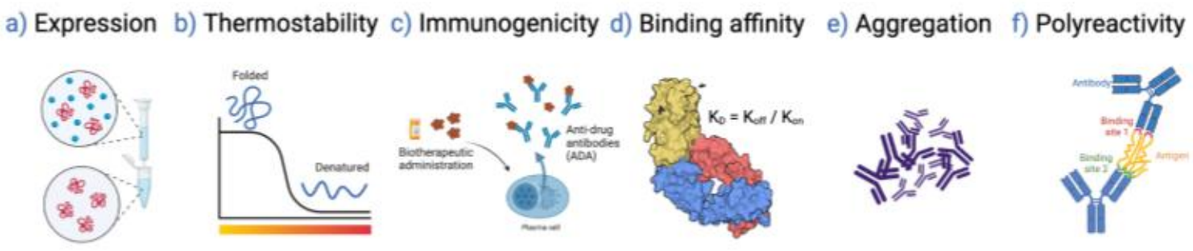
1 Background

Antibody therapeutics must have a particular set of fitness properties



Six classes of biophysical data relevant to antibody development are (a) expression, (b) thermostability, (c) immunogenicity, (d) binding affinity, (e) aggregation, and (f) polyreactivity

Do deep learning methods capture properties of therapeutic antibodies?

We seek to answer this question by:

1. Creating a collection of datasets of antibody therapeutic data
2. Benchmark representative AI model likelihoods for fitness prediction

2 Dataset curation

We curated publicly available antibody fitness datasets

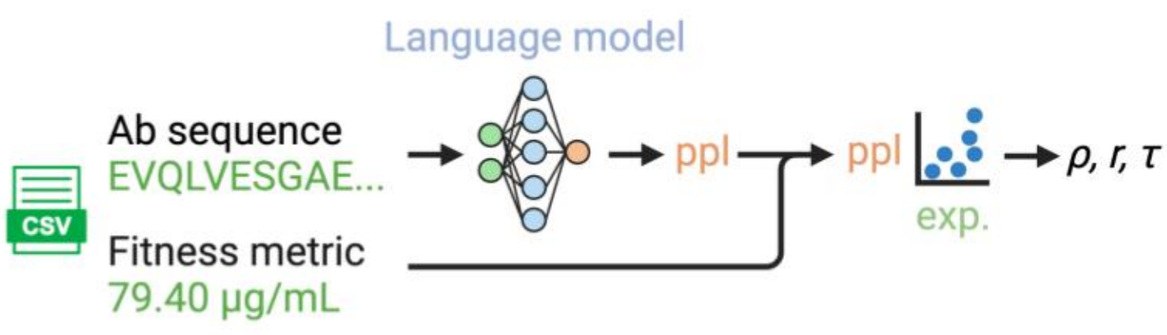
Number of datapoints

Dataset: Antibody	Exp. (μg/mL)	T_m (°C)	Imm. (% ADA)	Binding (nM)	Agg. (Wv shift)	Poly. (min)
GSK CA1	34	34	-	29	-	-
GSK CA2	25	22	-	22	-	-
GSK CA3	11	8	-	11	-	-
GSK CA4	24	24	-	19	-	-
Hie C143	-	2	-	16	-	-
Hie mAb114	-	7	-	20	-	-
Hie mAb114 UCA	-	2	-	-	-	-
Hie MEDI8852	-	2	-	15	-	-
Hie MEDI8852 UCA	-	6	-	20	-	-
Hie REGN10987	-	8	-	13	-	-
Hie S309	-	10	-	19	-	-
Koenig G6	4275	-	-	4275	-	-
Prihoda imm	-	-	217	-	-	-
Rosace Adalimumab	-	14	-	14	-	14
Rosace CD3022	-	6	-	6	-	6
Rosace Golimumab	-	5	-	5	-	5
Shane. Trast. multi	-	-	-	24	-	-
Shane. Trast. zero	-	-	-	422	-	-
Warszawski D44	-	-	-	2049	-	-
Wittrup CST	274	137	-	-	822	411

This database includes mutational landscapes of 17 distinct antibody families and a total of 13,384 associated fitness metrics. Each sequence is mapped to at least one fitness label pertaining to the 6 fitness properties

3 Methods

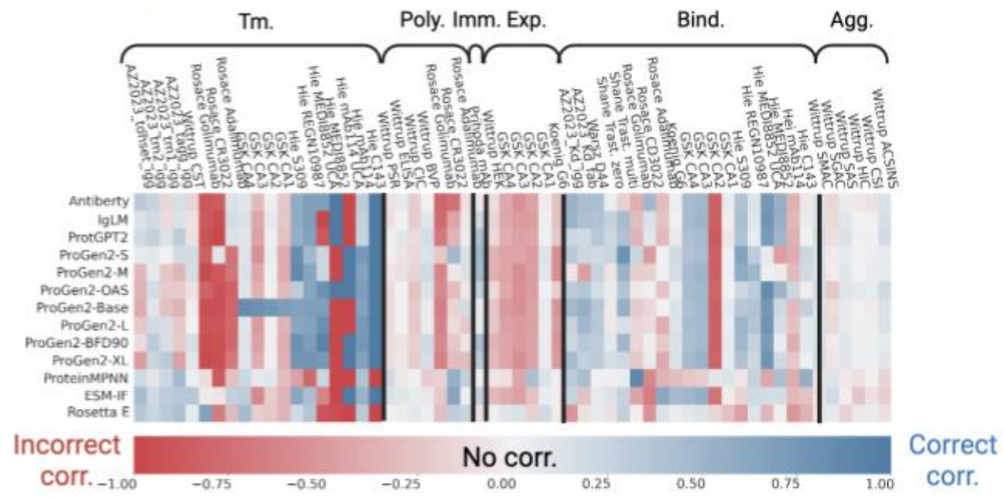
Pipeline for benchmarking protein language models



For each protein language model, we separately input the antibody **heavy** and **light sequences** to return two **perplexity scores**, and we tabulate the average perplexity between the two sequences. We calculate Spearman's (**ρ**), Pearson's (**r**) and Kendall tau's (**τ**) correlation coefficients between **perplexity** and **fitness**.

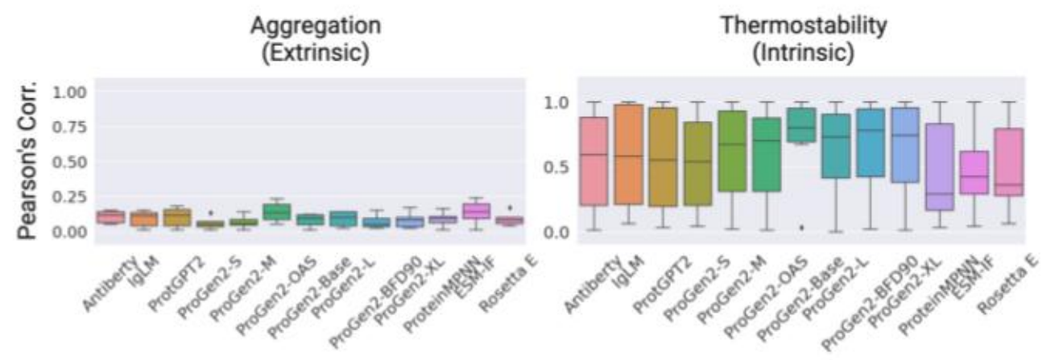
4 Results

A heat map of model-dataset correlations



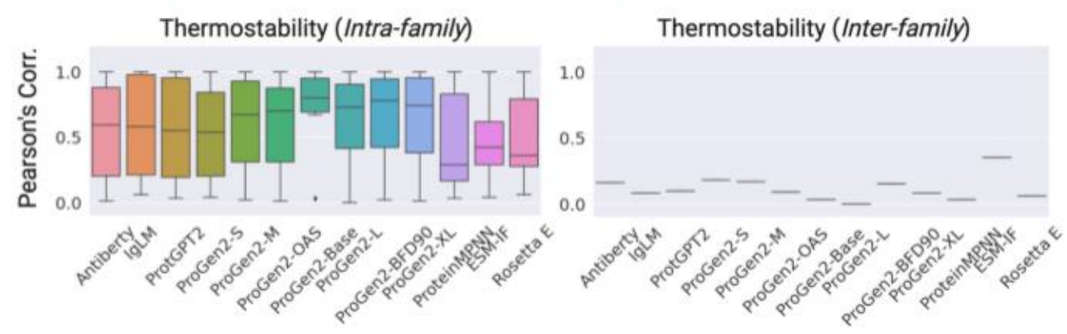
Summary of Pearson's correlations. Models generally perform best with thermostability and binding affinity datasets, but most struggle with aggregation propensity and polyreactivity.

Intrinsic biophysical properties are more accurately predicted than extrinsic



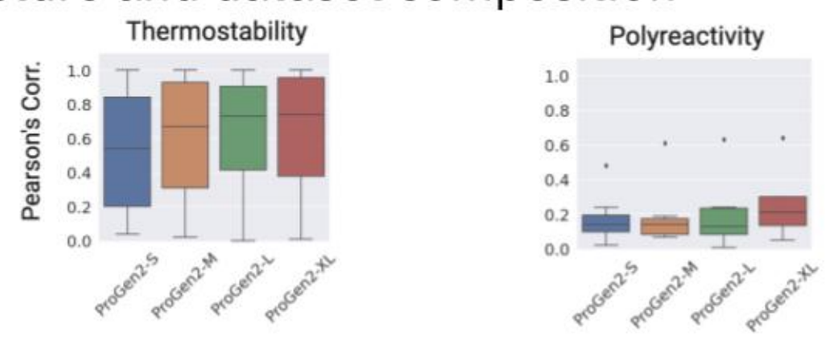
Intrinsic properties are impacted by inherent properties of the antibody. Extrinsic properties result from target biology and mechanisms of action.

Models are more accurate at distinguishing intra-family versus inter-family antibody sets



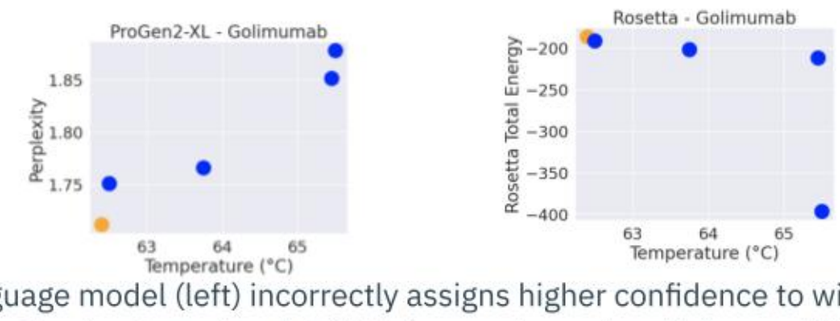
Intra-family antibodies are mutants originating from same wild type. Inter-family antibodies have different wild type origins.

Parameter size influences performance over architecture and dataset composition



Polyreactivity and thermostability performance correlations improve with parameter size for ProGen2 (151M, 764M, 2.7B, 6.4B parameters)

Some models favor evolutionary signal rather than physical fitness



The language model (left) incorrectly assigns higher confidence to wild type antibody. Physics-based model (right) correctly assigns higher stability to the mutants

5 Conclusion

FLAb is a living benchmark where we will continuously add models evaluated and available antibody data

No model was top performing across all six fitness classes